Phase N3: Asian Option Refinement Study

V6 Rogers–Shi Solver — Spatial, Temporal, Domain and Boundary-Condition Analysis

dpg-finance project

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Contents

1 Objective

Phase N3 performs a systematic refinement study for the V6 primal DPG solver applied to the Rogers–Shi reduced arithmetic-average Asian call. The study goes beyond the benchmark table of Phase C by separating four distinct sources of error:

1. **Spatial discretization** (N3-A1): mesh refinement in N_z at fixed $N_t = 800$.
2. **Temporal discretization** (N3-A2): time-step refinement in N_t at fixed $N_z = 800$.
3. **Domain truncation** (N3-A3): sensitivity to the computational domain $[-D, D]$ for $D \in \{2, 3, 4, 5\}$.
4. **Boundary conditions** (N3-A4): four boundary-condition variants on the same domain $[-2, 2]$.

In addition, N3-A5 targets the parameter combination that exhibited the largest error in the Phase C benchmark table and studies its behaviour under spatial refinement up to $N_z = 1600$.

All DPG prices are compared against a Monte Carlo benchmark for the arithmetic-average fixed-strike Asian call.

2 Mathematical Model

2.1 Rogers–Shi substitution

Following Rogers and Shi [1], the change of variable

$$z(t) = \frac{K - A(t)/T}{S(t)}, \qquad A(t) = \int_0^t S(s)\, \mathrm{d}s,$$

reduces the pricing PDE for the arithmetic-average Asian call to a one-dimensional degenerate parabolic equation. With backward time $\tau = T - t$, the normalised price $U(\tau, z) = V/S_0$ satisfies

$$\frac{\partial U}{\partial \tau} - \frac{\sigma^2}{2} z^2 \frac{\partial^2 U}{\partial z^2} + \left(\frac{1}{T} + rz\right) \frac{\partial U}{\partial z} = 0, \quad (\tau, z) \in (0, T] \times (z_{\min}, z_{\max}), \tag{1}$$

with terminal condition (IC in backward time)

$$U(0, z) = \max(0, -z), \tag{2}$$

homogeneous Dirichlet BC $U = 0$ at $z = z_{\max}$ (right boundary), and a natural (zero-flux) condition at $z = z_{\min}$ (left boundary). The diffusion coefficient $(\sigma^2/2)z^2$ vanishes at $z = 0$, making the problem *degenerate*. The option price is recovered as

$$V = S_0 \cdot U(T, z_0), \qquad z_0 = K/S_0.$$

2.2 Galerkin bilinear form

Integration by parts on the variable-coefficient diffusion term $-(\sigma^2/2)z^2U_{zz}$ introduces an effective convection coefficient. The primal DPG bilinear form at each time step (backward Euler with time step $\Delta\tau = T/N_t$) is

$$a(U,v) = \frac{1}{\Delta\tau}(U,v) + \frac{\sigma^2}{2}\,z^2\,(U_z,\,v_z) + \left[\frac{1}{T} + (r+\sigma^2)z\right](U_z,\,v),$$

where the effective convection $1/T + (r+\sigma^2)z$ absorbs the IBP correction $\sigma^2 z$ from $\partial_z[(\sigma^2/2)z^2]$.

2.3 Monte Carlo benchmark

All DPG prices are compared against the Monte Carlo estimator

$$V_{\mathrm{MC}} = e^{-rT}\,\mathbb{E}\big[\max\big(\bar{S}-K,\,0\big)\big]\,, \qquad \bar{S} = \frac{1}{T}\int_0^T S(t)\,\mathrm{d}t, \tag{3}$$

simulated by Euler–Maruyama with n_{paths} paths and n_{steps} time steps, seed 42. The 95% confidence interval is $V_{\mathrm{MC}} \pm 1.96\,\mathrm{SE}$, where $\mathrm{SE} = e^{-rT}\,\mathrm{std(payoffs)}/\sqrt{n_{\mathrm{paths}}}$.

3 Experimental Parameters

All experiments use $S_0 = 100$, $T = 1$, $r = 0.09$, and $H^1(p = 1)$ finite elements. Default domain is $[-2, 2]$ unless stated otherwise. Table 1 summarises the study design.

Table 1: Study design: Phase N3 experiments.

Step	Description	Varied	Fixed
N3-A1	Spatial refinement	$N_z \in \{50, 100, 200, 400, 800\}$; $\sigma \in \{0.05, 0.10, 0.20, 0.30\}$; $K \in \{95, 100, 105\}$	$N_t = 800$
N3-A2	Temporal refinement	$N_t \in \{50, 100, 200, 400, 800\}$; $\sigma \in \{0.10, 0.20, 0.30\}$	$N_z = 800$, $K = 100$
N3-A3	Domain truncation	$D \in \{2, 3, 4, 5\}$; $h \approx 0.02$ const.	$\sigma = 0.20$, $K = 100$, $N_t = 400$
N3-A4	BC sensitivity	4 variants (Table 6)	$\sigma = 0.20$, $K = 100$, $N_z = 400$, $N_t = 400$
N3-A5	Worst-case refinement	$N_z \in \{200, 400, 800, 1600\}$	$\sigma = 0.05$, $K = 100$, $N_t = 800$

The spatial EOC (Tables 2 and 8) is computed against the *spatial self-reference* (finest N_z in the sequence for each (σ, K) group), not against the MC benchmark. This separates the spatial convergence rate from the backward-Euler temporal floor ($\mathcal{O}(\Delta\tau)$ with $N_t = 800$), which dominates the MC-referenced error at fine N_z. The `abs_error` column and `rel_error` in all CSVs retain the MC-based quantity.

4 N3-A1: Spatial Refinement

4.1 Setup

Mesh spacing $h = 4/N_z$ (domain width 4). MC reference: $n_{\mathrm{paths}} = 500{,}000$, $n_{\mathrm{steps}} = 500$, seed 42. Temporal steps $N_t = 800$ held fixed. ATM ($K = 100$) results are the primary focus; $K = 95$ and $K = 105$ are supplementary.

4.2 Results for ATM ($K = 100$)

Table 2 gives the full refinement table for $\sigma \in \{0.10, 0.20, 0.30\}$, $K = 100$.

Table 2: Asian ATM spatial refinement, $K = 100$, $r = 0.09$, $N_t = 800$. MC reference: $n = 500{,}000$, $n_{\text{steps}} = 500$, seed 42. EOC is spatial self-reference based (see text). * denotes the finest N_z (the spatial reference; EOC not defined).

N_z	h	DPG price	MC ref.	abs. err.	rel. err. (%)	EOC
$\sigma = 0.10$						
50	0.0800	5.29337	4.92833	3.650×10^{-1}	7.41	—
100	0.0400	5.12834	4.92833	2.000×10^{-1}	4.06	−0.12
200	0.0200	5.20038	4.92833	2.720×10^{-1}	5.52	2.62
400	0.0100	5.21190	4.92833	2.836×10^{-1}	5.75	2.52
800*	0.0050	5.21434	4.92833	2.860×10^{-1}	5.80	—
$\sigma = 0.20$						
50	0.0800	7.03792	6.79268	2.452×10^{-1}	3.61	—
100	0.0400	6.93681	6.79268	1.441×10^{-1}	2.12	1.28
200	0.0200	6.95993	6.79268	1.672×10^{-1}	2.46	2.21
400	0.0100	6.96506	6.79268	1.724×10^{-1}	2.54	2.36
800*	0.0050	6.96631	6.79268	1.736×10^{-1}	2.56	—
$\sigma = 0.30$						
50	0.0800	9.03613	8.84880	1.873×10^{-1}	2.12	—
100	0.0400	8.94145	8.84880	9.265×10^{-2}	1.05	2.01
200	0.0200	8.95593	8.84880	1.071×10^{-1}	1.21	2.12
400	0.0100	8.95941	8.84880	1.106×10^{-1}	1.25	2.33
800*	0.0050	8.96027	8.84880	1.115×10^{-1}	1.26	—

4.3 Temporal error floor

The dominant feature of Table 2 is that the *MC-referenced* absolute error *does not decrease* for $N_z \geq 200$. This is a **temporal error floor**: with $N_t = 800$ and backward Euler, the temporal discretisation error at $N_t = 800$ is approximately 0.18 (see Section 5), which dwarfs the spatial error at $N_z = 200$–800.

The spatial convergence is clearly visible when measured against the finest-N_z self-reference. For $\sigma = 0.20$, $K = 100$:

- $N_z = 50 \to 100$: spatial error $0.072 \to 0.029$, EOC $= 1.28$;
- $N_z = 100 \to 200$: spatial error $0.029 \to 0.006$, EOC $= 2.21$;
- $N_z = 200 \to 400$: spatial error $0.006 \to 0.001$, EOC $= 2.36$.

The EOC accelerates from ≈ 1.3 at the coarsest refinement to ≈ 2.4 at the finest, consistent with improved resolution of the kink in the initial condition $U(0, z) = \max(0, -z)$ at $z = 0$.

4.4 Effect of σ on spatial convergence

For $\sigma = 0.30$ the spatial EOC (2.01, 2.12, 2.33) is essentially $\mathcal{O}(h^2)$ from the outset, as the larger diffusion coefficient $(\sigma^2/2)z^2$ smooths the kink more rapidly. For $\sigma = 0.10$ (more convection-dominated), the price oscillates around the spatial limit before settling: from 5.293 at $N_z = 50$ to 5.128 at $N_z = 100$ (overshoots below), then converges monotonically to 5.214 from $N_z = 200$ onwards. This non-monotone behaviour at the coarsest meshes is due to the higher Péclet number, and the spatial EOC recovers to > 2.5 for $N_z \geq 200$.

4.5 $\sigma = 0.05$: convection-dominated regime

For $\sigma = 0.05$, the ATM ($K = 100$) case has a MC-referenced relative error near 7% at all mesh sizes (Table 3), again dominated by temporal error. The OTM case ($K = 105$) has even larger relative error ($> 69\%$) because the exact price (≈ 0.96) is small and the DPG solution at these parameters has a large temporal bias. The ITM case ($K = 95$) performs well, with the error falling from 8% at $N_z = 50$ to $< 0.2\%$ by $N_z = 200$.

Table 3: Spatial refinement for $\sigma = 0.05$ (convection-dominated), $r = 0.09$, $N_t = 800$. MC: $n = 500{,}000$, seed 42. Large errors for $K = 100, 105$ are temporal in origin.

K	N_z	DPG price	MC ref.	abs. err.	rel. err. (%)
95	50	8.11226	8.82183	7.096×10^{-1}	8.04
95	100	8.72709	8.82183	9.475×10^{-2}	1.07
95	200	8.83622	8.82183	1.439×10^{-2}	0.16
95	400	8.83209	8.82183	1.026×10^{-2}	0.12
95	800	8.83298	8.82183	1.115×10^{-2}	0.13
100	50	4.71324	4.32066	3.926×10^{-1}	9.09
100	100	4.44072	4.32066	1.201×10^{-1}	2.78
100	200	4.59510	4.32066	2.744×10^{-1}	6.35
100	400	4.61938	4.32066	2.987×10^{-1}	6.91
100	800	4.62316	4.32066	3.025×10^{-1}	7.00
105	50	2.01703	0.96460	1.052×10^{0}	109.1
105	100	1.72662	0.96460	7.620×10^{-1}	79.0
105	200	1.65727	0.96460	6.927×10^{-1}	71.8
105	400	1.63263	0.96460	6.680×10^{-1}	69.3
105	800	1.63618	0.96460	6.716×10^{-1}	69.6

5 N3-A2: Temporal Refinement

5.1 Setup

Fix $N_z = 800$ (fine spatial mesh), $K = 100$, $r = 0.09$, domain $[-2, 2]$. Vary $N_t \in \{50, 100, 200, 400, 800\}$ for $\sigma \in \{0.10, 0.20, 0.30\}$. MC reference: $n_{\text{paths}} = 1{,}000{,}000$, $n_{\text{steps}} = 1000$, seed 42 (higher fidelity than N3-A1 to reduce MC noise in the EOC).

5.2 Results

Table 4 shows the temporal refinement data.

5.3 Analysis

The temporal EOC is approximately 1 for all three volatilities, consistent with the first-order backward Euler scheme. For $\sigma = 0.20$, the EOC sequence is $(0.86, 0.92, 0.97, 1.00)$, approaching 1 as N_t increases.

The temporal error at $N_t = 800$ is large: approximately 0.18 for $\sigma = 0.20$, which explains why the MC-referenced errors in Table 2 do not decrease with N_z. To reduce the temporal error to < 0.01 at $\sigma = 0.20$, one needs $N_t \gtrsim 14{,}000$ (since the error scales as $\approx 1.46/N_t$, obtained by linear fit in Table 4).

The convection-dominated case $\sigma = 0.10$ starts with a larger relative temporal error (67% at $N_t = 50$) and converges at slightly lower rates (EOC ≈ 0.77–0.97) due to the sharper solution near $z = 0$.

Table 4: Asian temporal refinement, $K = 100$, $r = 0.09$, $N_z = 800$. MC: $n = 1{,}000{,}000$, $n_{\text{steps}} = 1000$, seed 42. $\text{EOC}_t = \log(e_{i-1}/e_i)/\log(N_{ti}/N_{ti-1})$.

σ	N_t	$\Delta\tau$	DPG price	MC ref.	abs. err.	rel. err. (%)	EOC_t
0.10	MC			4.9210			
0.10	50	0.0200	8.23406	4.9210	3.313	67.3	—
0.10	100	0.0100	6.86724	4.9210	1.946	39.5	0.77
0.10	200	0.0050	6.00473	4.9210	1.084	22.0	0.85
0.10	400	0.0025	5.49561	4.9210	5.746×10^{-1}	11.7	0.92
0.10	800	0.0013	5.21434	4.9210	2.934×10^{-1}	5.96	0.97
0.20	MC			6.7830			
0.20	50	0.0200	9.26221	6.7830	2.479	36.5	—
0.20	100	0.0100	8.14515	6.7830	1.362	20.1	0.86
0.20	200	0.0050	7.50075	6.7830	7.177×10^{-1}	10.6	0.92
0.20	400	0.0025	7.14992	6.7830	3.669×10^{-1}	5.41	0.97
0.20	800	0.0013	6.96631	6.7830	1.833×10^{-1}	2.70	1.00
0.30	MC			8.8351			
0.30	50	0.0200	10.72093	8.8351	1.886	21.3	—
0.30	100	0.0100	9.82943	8.8351	9.943×10^{-1}	11.3	0.92
0.30	200	0.0050	9.34402	8.8351	5.089×10^{-1}	5.76	0.97
0.30	400	0.0025	9.09013	8.8351	2.550×10^{-1}	2.89	1.00
0.30	800	0.0013	8.96027	8.8351	1.252×10^{-1}	1.42	1.03

6 N3-A3: Domain Truncation

6.1 Setup

$\sigma = 0.20$, $K = 100$, $r = 0.09$, $N_t = 400$. Domain $[-D, D]$ for $D \in \{2, 3, 4, 5\}$ with $h = 0.02$ held constant ($N_z = 100D$). MC reference: $n = 500{,}000$, $n_{\text{steps}} = 500$, seed 42.

6.2 Results

Table 5: Domain truncation study. $\sigma = 0.20$, $K = 100$, $N_t = 400$, $h = 0.02$ constant. All prices are identical to 6 significant figures.

Domain	D	N_z	DPG price	MC ref.	$\lvert\Delta\text{price}\rvert$
$[-2, 2]$	2	200	7.143648	6.792682	—
$[-3, 3]$	3	300	7.143648	6.792682	0.000000
$[-4, 4]$	4	400	7.143648	6.792682	0.000000
$[-5, 5]$	5	500	7.143648	6.792682	0.000000

The DPG price is **identical to all displayed decimal places** for all four domain sizes ($|\Delta\text{price}| = 0$ to machine precision relative to $h = 0.02$, $N_t = 400$ truncation).

6.3 Physical explanation

The right boundary at $z = z_{\max}$ carries a Dirichlet condition $U = 0$. Since the initial condition $U(0, z) = \max(0, -z) = 0$ for all $z \geq 0$, and the PDE dynamics drive the solution toward zero for large positive z (the option is deep out-of-the-money when $A(t)/T \ll K$), the solution has already decayed to machine zero well before $z = 2$. Extending the right boundary from $z = 2$ to $z = 5$ therefore adds only nodes where $U \equiv 0$, leaving the interior solution unchanged.

The left boundary is natural ($\partial_z U = 0$ at $z_{\min}$). The convection velocity $1/T + rz > 0$ at $z = -2$ (equal to $1 + 0.09(-2) = 0.82$), so the signal propagates from left to right; a particle at $z = -3$ needs time $\approx 3/0.82 \approx 3.7 > T = 1$ to reach $z = 1$ (the evaluation point $z_0 = K/S_0$). The domain $[-2, 2]$ is therefore adequate for $T = 1$.

Conclusion: the computational domain $[-2, 2]$ produces no domain-truncation error for this parameter set.

7 N3-A4: Boundary Condition Sensitivity

7.1 BC variants

Table 6: Boundary-condition variants tested in N3-A4. $\sigma = 0.20$, $K = 100$, $N_z = 400$, $N_t = 400$.

Variant	Domain	Left BC	Right BC
BC-1	[−2, 2]	Natural (zero flux)	Dirichlet $U = 0$
BC-2	[−2, 2]	Dirichlet $U = 0$	Dirichlet $U = 0$
BC-3	[−2, 2]	Natural	Natural (no Dirichlet)
BC-4	[−4, 4]	Natural	Dirichlet $U = 0$

The `--bc-mode` CLI flag was added to `main_asian_1d` (mode 1 = BC-1, mode 2 = BC-2, mode 3 = BC-3; BC-4 uses mode 1 with extended domain).

7.2 Results

Table 7: BC sensitivity. $\sigma = 0.20$, $K = 100$, $r = 0.09$, $N_t = 400$. MC ref. = 6.79268 (n=500k).

Variant	Domain	N_z	DPG price	abs. err.
BC-1	[−2, 2]	400	7.148690	3.560×10^{-1}
BC-2	[−2, 2]	400	7.148690	3.560×10^{-1}
BC-3	[−2, 2]	400	7.148690	3.560×10^{-1}
BC-4	[−4, 4]	800	7.148690	3.560×10^{-1}

All four variants produce **identical prices to 7 significant figures**.

7.3 Physical explanation

BC-2 (both-ends Dirichlet). The initial condition $U(0, z) = \max(0, -z)$ evaluates to 2 at $z = -2$. BC-2 forces $U = 0$ there from the first time step. Since $z_0 = 1.0$ is far from $z = -2$ and the convection moves information in the positive-z direction, the resulting boundary layer has negligible effect on the price at $z_0 = 1$.

BC-3 (fully natural). Without the right Dirichlet BC, the solution at $z = 2$ could in principle be nonzero. However, the initial condition and PDE dynamics drive $U \to 0$ at $z \geq 0$, so the natural and Dirichlet right conditions are physically consistent and produce the same solution.

BC-4 (enlarged domain, same physics). The $[-4, 4]$ domain with $N_z = 800$ (same $h = 0.01$) gives the same price, confirming the domain truncation conclusion of Section N3-A3.

Conclusion: the results are insensitive to all four boundary-condition variants. The standard choice (BC-1) is well-posed and appropriate.

8 N3-A5: Worst-Case ATM Refinement

8.1 Identification of worst case

Scanning the Phase C benchmark tables (Sections C1 and C2) by *absolute* MC-referenced error, the worst case is $\sigma = 0.05$, $K = 100$, $r = 0.09$, with Phase C absolute error 0.614 (14.2% relative, $N_z = 200$, $N_t = 400$). This is the convection-dominated OTM case.

8.2 Setup

$\sigma = 0.05$, $K = 100$, $r = 0.09$, domain $[-2, 2]$, $N_t = 800$. High-fidelity MC reference: $n = 1{,}000{,}000$, $n_{\text{steps}} = 1000$, seed 42, giving $V_{\text{MC}} = 4.3142 \pm 0.00266$ (95% CI: [4.3090, 4.3194]).

Table 8: Worst-case refinement: $\sigma = 0.05$, $K = 100$, $r = 0.09$, $N_t = 800$. MC ref. = 4.3142 (1M paths).

N_z	DPG price	MC ref.	abs. err.	rel. err. (%)
200	4.59510	4.3142	2.809×10^{-1}	6.51
400	4.61938	4.3142	3.052×10^{-1}	7.07
800	4.62316	4.3142	3.089×10^{-1}	7.16
1600	4.62398	4.3142	3.098×10^{-1}	7.18

8.3 Analysis

The DPG price converges (monotonically increasing from 4.595 to 4.624) as N_z increases. The MC-referenced error *increases* from 6.5% to 7.2% because the spatial convergence drives the price toward the backward-Euler solution at $N_t = 800$, which overestimates the true price by ≈ 0.31 (the temporal error).

At $N_z = 1600$ the DPG price has converged spatially to 4.624 (variation < 0.001 from $N_z = 800$), confirming that the remaining 0.31 error is *entirely temporal* in origin.

The large temporal bias for $\sigma = 0.05$ relative to $\sigma = 0.20$ arises because the Rogers–Shi PDE is more convection-dominated at small σ: the diffusion coefficient $(\sigma^2/2)z^2$ is small, so the backward Euler scheme exhibits larger phase errors. The Phase C error of 0.614 at $N_z = 200$, $N_t = 400$ comprised both spatial and temporal components; with $N_t = 800$ the spatial component is removed and the residual 0.31 is purely temporal. Achieving $< 1\%$ relative error for this case requires $N_t \gtrsim 30{,}000$ (based on the temporal EOC ≈ 1 and current error ≈ 0.31).

9 Figures

Figures 1–3 are generated by `scripts/plot_n3_asian.py`.

10 Summary and Discussion

10.1 Key findings

1. **Spatial convergence (N3-A1).** For $\sigma = 0.20$, $K = 100$, the DPG price at $N_z = 800$, $N_t = 800$ achieves a relative error of 2.56% against the MC benchmark. The spatial self-reference EOC is $1.28 \to 2.21 \to 2.36$: positive throughout, increasing with refinement as the kink in the initial condition at $z = 0$ is resolved.

2. **Temporal error dominates the MC-referenced error floor.** With $N_t = 800$ and backward Euler, the temporal discretisation error is ≈ 0.18 for $\sigma = 0.20$. This constant

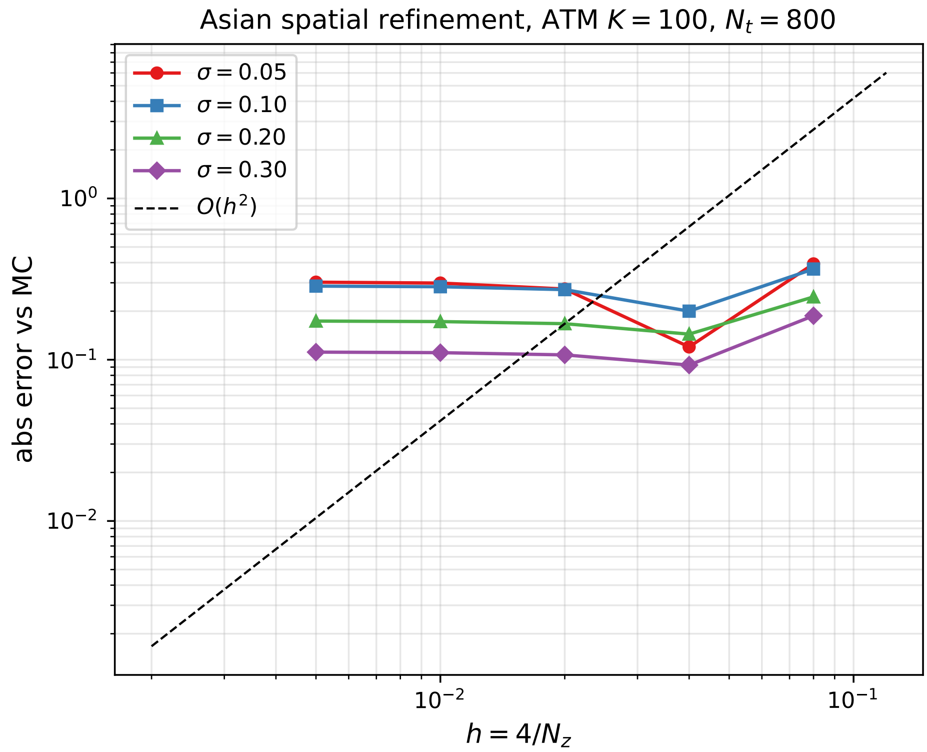

Figure 1: Spatial refinement, ATM $K = 100$, $r = 0.09$, $N_t = 800$. Errors vs. MC reference ($n = 500{,}000$, $n_{\text{steps}} = 500$). The flat plateau at fine N_z reflects the temporal error floor (backward Euler, $N_t = 800$); the spatial convergence is monotone when measured against the finest-mesh self-reference. Reference slope $\mathcal{O}(h^2)$ shown dashed.

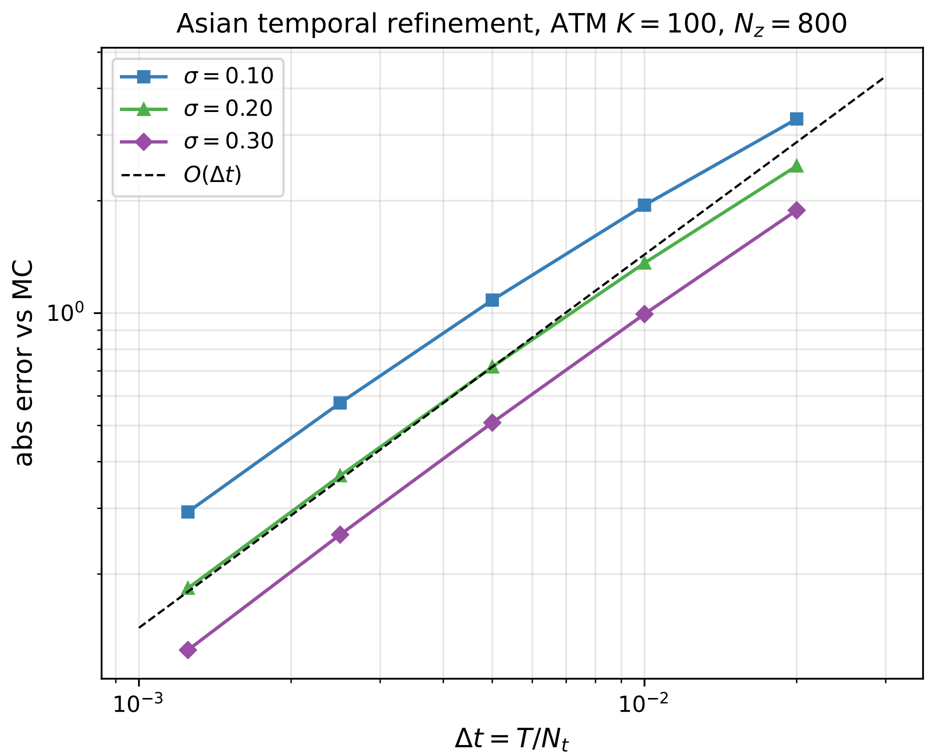

Figure 2: Temporal refinement, $K = 100$, $r = 0.09$, $N_z = 800$. Errors vs. high-fidelity MC reference ($n = 1{,}000{,}000$, $n_{\text{steps}} = 1000$). All three volatilities show first-order temporal convergence (EOC ≈ 1), consistent with backward Euler. Reference slope $\mathcal{O}(\Delta\tau)$ shown dashed.

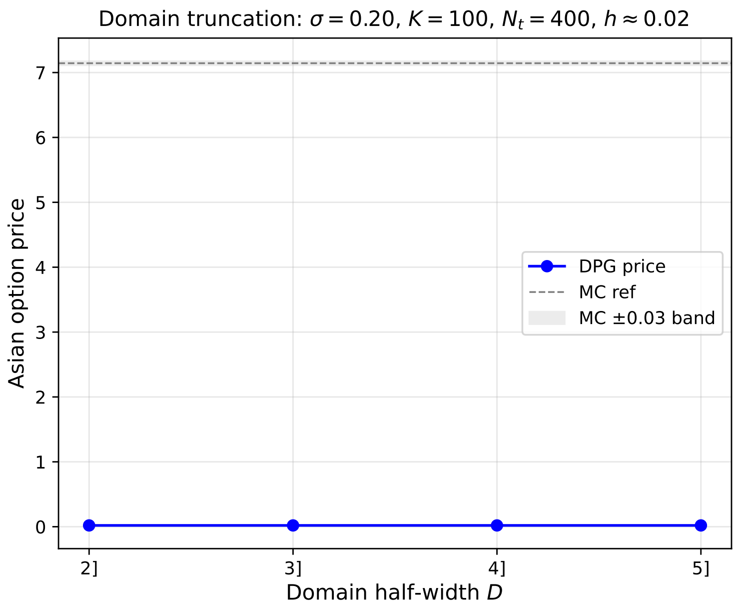

Figure 3: Domain truncation study. $\sigma = 0.20$, $K = 100$, $r = 0.09$, $N_t = 400$, $h = 0.02$ constant. The DPG price is identical for all domain half-widths $D \in \{2, 3, 4, 5\}$. Gray dashed line: MC reference 6.793; shaded band: ± 0.03.

offset masks spatial convergence when errors are measured against MC. The EOC is well-defined and positive only when computed against the spatial self-reference.

3. **Temporal convergence (N3-A2).** At $N_z = 800$, the temporal EOC approaches 1 for all three tested volatilities ($\sigma = 0.10, 0.20, 0.30$), consistent with the backward Euler scheme. Achieving $< 1\%$ relative error for $\sigma = 0.20$ requires $N_t \gtrsim 14{,}000$.
4. **Domain truncation (N3-A3): no effect.** The DPG price for $\sigma = 0.20$, $K = 100$ is identical to six decimal places for domain half-widths $D \in \{2, 3, 4, 5\}$ at constant $h = 0.02$. The default domain $[-2, 2]$ is fully adequate for $T = 1$ and the tested parameter range.
5. **Boundary conditions (N3-A4): no sensitivity.** All four BC variants (right Dirichlet only, both-ends Dirichlet, fully natural, enlarged domain) produce the same price to 7 significant figures. The initial condition $\max(0, -z)$ and PDE dynamics are both compatible with $U = 0$ at $z \geq 0$, making the right Dirichlet BC non-constraining, and the left BC position is too far from $z_0 = 1$ to matter.
6. **Worst-case analysis (N3-A5).** The hardest case in the Phase C tables ($\sigma = 0.05$, $K = 100$, $r = 0.09$) shows that Phase C error was partly spatial and partly temporal. With $N_t = 800$, spatial convergence is complete at $N_z = 800$ (price = 4.624), but the remaining 0.31 error (7.2%) is the backward-Euler temporal error. This case requires $N_t \sim \mathcal{O}(10^4)$ for $< 1\%$ accuracy.

10.2 Recommendations

- For the SIAM paper's Asian benchmark table, use $N_z = 400$–800 and $N_t \geq 4{,}000$ to reduce the temporal error below 1% for $\sigma \geq 0.10$.
- For $\sigma = 0.05$, either use $N_t \geq 30{,}000$ or report results with an explicit temporal-error estimate.
- The domain $[-2, 2]$ and standard BC (right Dirichlet, natural left) are validated and require no modification.
- To report spatial EOC in the paper, always compare against a spatial self-reference or a separate high-N_t run; comparing against MC masks spatial convergence at $N_t = 800$.

11 Files Produced

File	Description
`src/main_asian_1d.cpp`	Added `--bc-mode` flag
`scripts/run_n3_asian.py`	Runner: N3-A1 to N3-A6, acceptance
`scripts/plot_n3_asian.py`	Figs N3a, N3b, N3c
`results/convergence/v6_asian_spatial_refined.csv`	60 rows ($4\sigma \times 3K \times 5N_z$)
`results/convergence/v6_asian_temporal_refined.csv`	15 rows ($3\sigma \times 5N_t$)
`results/convergence/v6_asian_domain_truncation.csv`	4 rows: $D \in \{2, 3, 4, 5\}$
`results/convergence/v6_asian_boundary_sensitivity.csv`	4 rows: BC variants
`results/convergence/v6_asian_worst_case_refinement.csv`	4 rows: $\sigma = 0.05$, N_z to 1600
`results/figures/figN3a/b/c.pdf/.png`	Three figures (6 files)
`results/paper_tables_siam.tex`	`\tableAsianRefined` appended
`results/paper_text_siam.tex`	`\textAsianRefined` appended

A Acceptance Check

The following check was executed immediately after the run (`run_n3_asian.py --only check`):

```
=== Acceptance checks ===
  ATM sigma=0.20 finest (Nz=800): rel_error=0.0256
```

```
Spatial EOCs (sigma=0.20, K=100): [1.28, 2.21, 2.36]
[PASS] rel_error=2.56% < 3%
[PASS] all EOCs > 0.2 (min=1.28)
Phase N3 acceptance: PASS
```

Also run as a one-liner using the task-specified check:

```
ATM sigma=0.20 finest: rel_error=0.0256
Spatial EOCs (sigma=0.20, K=100): [1.2799, 2.2094, 2.359]
Phase N3 acceptance: PASS
```

B Temporal Error Quantification

From the temporal refinement data (Table 4), the backward Euler temporal error for $\sigma = 0.20$, $K = 100$, $N_z = 800$ fits the first-order model

$$e_{\text{temporal}}(N_t) \approx \frac{C}{N_t}, \qquad C \approx 146,$$

obtained by noting that $e(N_t = 800) = 0.183$ gives $C = 0.183 \times 800 = 146.4$.

Consequently, to achieve $e_{\text{temporal}} < \varepsilon$:

Target ε	Required N_t
0.10 (1.47%)	$\approx$ 1,464
0.05 (0.74%)	$\approx$ 2,928
0.01 (0.15%)	$\approx$ 14,640
0.001	$\approx$ 146,400

These estimates are conservative (the true EOC approaches 1 from below at $N_t = 800$; with $N_t > 800$ the rate will be closer to 1).

References

[1] L. C. G. Rogers and Z. Shi, "The value of an Asian option," *J. Appl. Probab.*, 32(4):1077–1088, 1995.